OAVS TGT - Current Affairs

Part 6: SCIENCE & ENVIRONMENT

Expected questions with answers & explanations
based on PYQ pattern analysis (2019, 2022, 2023, 2024)

Science, space, environment and health developments roughly Sep 2025 - Oct 2026, plus static GK. Facts verified as of 28 Oct 2026.

37 MCQs | 3 topics | star = very high probability

PYQ Pattern Analysis - Science & Environment

Syllabus: *Current Affairs (30 marks) - recent developments in Sports, Awards, Politics, Finance & Banking, International happenings, etc.* Across the five papers analysed, science, space, environment and health items contributed 2-5 questions per paper. The recurring question types are:

Question type	Typical example from PYQs	How to prepare
Space & ISRO	HySIS hyperspectral satellite (Nov 2018); GARUDA grid; space-agency MoU 2022	Latest launches, rocket names, dates, Gaganyaan
Environment & wildlife	TX2 award - Pench (2023); National Clean Air Programme; Railways net zero 2030	Ramsar/World Heritage counts, tigers, cheetahs, NDC
Health & reports	WHO air quality 2022 (99%); AMR; World Health Day theme	Disease outbreaks, WHO days and themes
Technology	5G launch 1 Oct 2022; CSIR inventions	AI summit, semiconductors, digital initiatives

Strategy: most questions ask which mission/site/species/programme, with counts and dates. Prioritise starred questions and the Rapid-Revision table at the end.

Contents

1. Space & ISRO (10 MCQs)
2. Environment, Wildlife & Heritage (16 MCQs)
3. Technology, AI & Health (11 MCQs)
4. Rapid-revision table

1. Space & ISRO

Why it matters: ISRO launches are a permanent fixture (HySIS satellite 2018, space-agency MoU 2022, Geospatial Hackathon 2023). Learn the mission name, rocket, date and purpose.

KEY FACTS

• Gaganyaan (India's first human spaceflight programme): first uncrewed test flight G1 on the human-rated LVM3 (HLVM3) from Sriharikota carrying the half-humanoid robot Vyommitra - ISRO Chairman Dr V. Narayanan said on 24 Oct 2026 it will fly in about two months (around December 2026). G2 follows; the crewed flight is expected in 2027.

• NISAR (NASA-ISRO Synthetic Aperture Radar): launched 30 Jul 2025 on GSLV-F16 from Sriharikota - a joint ISRO-NASA Earth-observation mission.

• Axiom-4 (June-July 2025): Group Captain Shubhanshu Shukla became the first Indian to stay and work on the International Space Station (18 days).

• SpaDeX (launched 30 Dec 2024): India docked two satellites in space on 16 Jan 2025 - the 4th country after the USA, Russia and China.

• GSLV-F17 / EOS-05 (4 Sep 2026): successful launch of the Earth-observation satellite - ISRO's first mission after a seven-month gap. Upcoming: Chandrayaan-4 (lunar sample return), Chandrayaan-5 / LUPEX with Japan (Moon's south pole ice), a Venus orbiter.

• NASA's Artemis II crewed lunar flyby ended with splashdown on 11 Apr 2026; a crewed lunar landing is planned for 2028. PYQ anchors: ISRO's HySIS hyperspectral imaging satellite (Nov 2018); GARUDA - India's national grid computing initiative.

Practice questions

Q1. ★ ISRO's first uncrewed Gaganyaan test flight is called:

(A) Gaganyaan-X
(B) G0
(C) G1
(D) G2

Answer: (C) G1. Gaganyaan-1 (G1), using the HLVM3 rocket; due around December 2026.

Q2. ★ Which robot will fly on the Gaganyaan G1 uncrewed mission?

(A) Vyommitra (half-humanoid)
(B) Vikram
(C) Mitra-1
(D) Rover Pragyan

Answer: (A) Vyommitra (half-humanoid). Vyommitra - a half-humanoid robot standing in for an astronaut.

Q3. Who is the Chairman of ISRO?

(A) A. S. Kiran Kumar
(B) K. Sivan
(C) S. Somanath
(D) V. Narayanan

Answer: (D) V. Narayanan. Dr V. Narayanan (since January 2025).

Q4. The launch vehicle used for Gaganyaan's human-rated missions is:

(A) PSLV-XL
(B) HLVM3 (human-rated LVM3)
(C) SSLV
(D) GSLV Mk II

Answer: (B) HLVM3 (human-rated LVM3). A human-rated version of the LVM3.

Q5. ★ NISAR, launched on 30 July 2025, is a joint Earth-observation mission of ISRO and:

(A) JAXA
(B) ESA
(C) NASA
(D) Roscosmos

Answer: (C) NASA. NASA - launched on GSLV-F16.

Q6. ★ Who became the first Indian to stay and work on the International Space Station (Axiom-4, 2025)?

(A) Shubhanshu Shukla
(B) Sunita Williams
(C) Prasanth Nair
(D) Rakesh Sharma

Answer: (A) Shubhanshu Shukla. Group Captain Shubhanshu Shukla - an 18-day stay.

Q7. ★ India became the _____ country to achieve satellite docking in space with SpaDeX (January 2025).

(A) second
(B) third
(C) fourth
(D) fifth

Answer: (C) fourth. 4th, after the USA, Russia and China.

Q8. Which satellite did ISRO launch on GSLV-F17 on 4 September 2026?

(A) NISAR
(B) Cartosat-3B
(C) NVS-03
(D) EOS-05

Answer: (D) EOS-05. EOS-05 - the first ISRO launch after a seven-month gap.

Q9. Chandrayaan-4 is planned as India's first:

(A) Solar mission
(B) Lunar sample-return mission
(C) Mars landing
(D) Venus flyby

Answer: (B) Lunar sample-return mission. A lunar sample-return mission; Chandrayaan-5 (LUPEX) is a joint mission with Japan.

Q10. In November 2018, ISRO launched the HySIS satellite for:

(A) Weather forecasting
(B) Communication
(C) Hyper-spectral imaging
(D) Navigation

Answer: (C) Hyper-spectral imaging. HySIS = Hyperspectral Imaging Satellite (asked in OAVS TGT English 2019).

2. Environment, Wildlife & Heritage

Why it matters: PYQs asked 'TX2 award - Pench Tiger Reserve', 'National Clean Air Programme - which ministry', 'Railways net zero by 2030', and air-quality reports. Learn counts (Ramsar, tigers, cheetahs, World Heritage), new additions and India's climate targets.

KEY FACTS

• Ramsar sites: India's tally reached 103 on 15 Oct 2026 (announced by Environment Minister Bhupender Yadav) with Sukhna Wetland (Chandigarh; Ramsar Site No. 2602) and Sauj Jheel (Mainpuri, Uttar Pradesh; No. 2603). Ramsar Convention = wetlands treaty (Iran, 1971).

• UNESCO World Heritage: Ancient Buddhist Site at Sarnath inscribed at the 48th session of the World Heritage Committee (Busan, South Korea, July 2026) - India's 45th site (criteria iii and vi); India ranks 6th globally and 2nd in Asia-Pacific. Components: Dhamekh Stupa, Chaukhandi Stupa, Dharmarajika Stupa, Mulagandhakuti Vihara, Ashokan Pillar (Lion Capital = National Emblem). India's 44th: Maratha Military Landscapes (2025).

• Tigers: All India Tiger Estimation 2026 (sixth cycle) began in January 2026; final report expected in 2027; the 2022 estimate was 3,682 tigers (about 75% of the world's wild tigers). India has 58 tiger reserves; Project Tiger launched 1973; NTCA formed 2005. International Tiger Day = 29 July (2026 theme: 'Securing the Future of Tigers with Indigenous Peoples and Local Communities at the Heart').

• Cheetahs (Project Cheetah): 52 cheetahs in India (Sep 2026) - 49 at Kuno National Park and 3 at Gandhi Sagar Wildlife Sanctuary (both Madhya Pradesh); first batch from Namibia (Sept 2022), 12 from South Africa (Feb 2023), 9 from Botswana (Feb 2026); 32 India-born. Next sites: Nauradehi (MP) and Banni grasslands (Gujarat).

• India's updated NDC (2031-35), submitted to the UNFCCC in April 2026: cut emission intensity of GDP by at least 47% from 2005 levels by 2035; about 60% of installed power capacity from non-fossil sources; carbon sink of 3.5-4.0 billion tonnes CO2e. India's net-zero target: 2070. Indian Railways: net zero by 2030.

• PYQ anchors: TX2 Award (tiger doubling) - Pench Tiger Reserve (Jan 2023); National Clean Air Programme - launched by the Ministry of Environment, Forest and Climate Change (2019); WHO Air Quality Database 2022 - 99% of people breathe unhealthy air.

Practice questions

Q11. ★ How many Ramsar sites does India have after the October 2026 additions?

(A) 108
(B) 103
(C) 100
(D) 98

Answer: (B) 103. 103 - Sukhna Wetland (Chandigarh) and Sauj Jheel (Uttar Pradesh) were added on 15 Oct 2026.

Q12. Which of the following wetlands was designated a Ramsar site in October 2026?

(A) Sukhna Wetland, Chandigarh
(B) Loktak Lake
(C) Wular Lake
(D) Chilika Lake

Answer: (A) Sukhna Wetland, Chandigarh. Sukhna Wetland (Site No. 2602); Loktak, Chilika and Wular were designated earlier.

Q13. Sauj Jheel, a new Ramsar site, is located in which state?

(A) Rajasthan
(B) Haryana
(C) Gujarat
(D) Uttar Pradesh

Answer: (D) Uttar Pradesh. Mainpuri district, Uttar Pradesh.

Q14. ★ Which site was inscribed as India's 45th UNESCO World Heritage Site in July 2026?

(A) Sanchi Stupa
(B) Ancient Buddhist Site at Sarnath
(C) Nalanda Mahavihara
(D) Bodh Gaya Complex

Answer: (B) Ancient Buddhist Site at Sarnath. Sarnath - at the 48th World Heritage Committee session in Busan, South Korea.

Q15. The 48th session of the UNESCO World Heritage Committee (July 2026) was held in:

(A) Paris
(B) New Delhi
(C) Busan, South Korea
(D) Bonn

Answer: (C) Busan, South Korea. Busan, South Korea (20-29 July 2026).

Q16. The Lion Capital of Sarnath's Ashokan Pillar is India's:

(A) National Emblem
(B) National Flag emblem only
(C) Supreme Court seal
(D) RBI logo

Answer: (A) National Emblem. India's National Emblem is adapted from the Lion Capital of Ashoka at Sarnath.

Q17. India has how many tiger reserves (as of 2026)?

(A) 62
(B) 58
(C) 55
(D) 53

Answer: (B) 58. 58 tiger reserves; Project Tiger launched in 1973.

Q18. ★ The 2022 tiger estimation counted how many tigers in India?

(A) 2,967
(B) 3,167
(C) 3,682
(D) 4,100

Answer: (C) 3,682. 3,682 (2018: 2,967). The 2026 estimation's report is expected in 2027.

Q19. International Tiger Day is observed on:

(A) 3 March
(B) 5 June
(C) 22 May
(D) 29 July

Answer: (D) 29 July. 29 July - decided at the 2010 St. Petersburg Tiger Summit.

Q20. ★ How many cheetahs are in India as of September 2026, mostly at Kuno National Park?

(A) 20
(B) 34
(C) 52
(D) 75

Answer: (C) 52. 52 - 49 at Kuno, 3 at Gandhi Sagar.

Q21. In February 2026, nine cheetahs were brought to India from:

(A) South Africa
(B) Kenya
(C) Namibia
(D) Botswana

Answer: (D) Botswana. Botswana - to improve genetic diversity.

Q22. Cheetahs were first reintroduced in India in September 2022 at which place?

(A) Gir National Park
(B) Kuno National Park
(C) Jim Corbett
(D) Ranthambore

Answer: (B) Kuno National Park. Kuno National Park, Sheopur (Madhya Pradesh) - eight cheetahs from Namibia.

Q23. ★ India's updated NDC for 2031-35 targets at least what cut in emission intensity of GDP from 2005 levels by 2035?

(A) 33%
(B) 45%
(C) 47%
(D) 55%

Answer: (C) 47%. At least 47%; about 60% non-fossil power capacity.

Q24. In which year does India aim to achieve net-zero emissions?

(A) 2050
(B) 2060
(C) 2070
(D) 2075

Answer: (C) 2070. 2070 (announced at COP26, 2021).

Q25. ★ The TX2 award for doubling tiger numbers was given in January 2023 to which tiger reserve (asked in OAVS TGT Social Studies 2023)?

(A) Bandipur
(B) Sundarbans
(C) Pench Tiger Reserve
(D) Kaziranga

Answer: (C) Pench Tiger Reserve. Pench Tiger Reserve (Madhya Pradesh).

Q26. The National Clean Air Programme (NCAP) was launched by the Ministry of:

(A) Environment, Forest and Climate Change
(B) Science and Technology
(C) Urban Affairs
(D) Health

Answer: (A) Environment, Forest and Climate Change. MoEFCC, in 2019 (asked in OAVS TGT English 2019).

3. Technology, AI & Health

Why it matters: The PYQs asked about 5G launch (1 Oct 2022), AMR full form, CSIR inventions and the WHO health-day themes. Expect AI, semiconductors and disease outbreaks in the next paper.

KEY FACTS

• India AI Impact Summit 2026 (Bharat Mandapam, New Delhi, from 16 Feb 2026): PM Modi unveiled the 'MANAV' vision (Moral and ethical systems, Accountable governance, National sovereignty, Accessible and inclusive AI, Valid and legitimate AI); India joined the US-led Pax Silica initiative on 20 Feb 2026; sovereign models unveiled under the IndiaAI Mission (Rs 10,000 crore): Sarvam AI, Gnani AI's Vachana, BharatGen's Param2; Reliance-Jio announced Rs 10 lakh crore investment.

• India Semiconductor Mission (ISM) 2.0 was announced in the Union Budget 2026-27.

• Nipah virus: two health workers were infected in West Bengal in late December 2025 (India's 7th documented outbreak, 3rd in West Bengal); WHO assessed the risk of wider spread as low. Nipah - carried by fruit bats; fatality 40-75%; no licensed vaccine; WHO priority pathogen.

• PYQ anchors: AMR = Anti-Microbial Resistance; 5G services launched in India on 1 Oct 2022 at the 6th India Mobile Congress; NHA implements Ayushman Bharat PM-JAY; WHO World Health Day 2021 theme: 'Building a fairer, healthier world'.

Practice questions

Q27. ★ The MANAV vision for ethical AI governance was unveiled by PM Modi at the:

(A) G20 Summit 2025
(B) BRICS Summit 2026
(C) World Economic Forum 2026
(D) India AI Impact Summit 2026

Answer: (D) India AI Impact Summit 2026. India AI Impact Summit, Bharat Mandapam, February 2026.

Q28. In the MANAV vision, the letter 'N' stands for:

(A) New technology
(B) National sovereignty
(C) Nano-science
(D) Network neutrality

Answer: (B) National sovereignty. National sovereignty ('Whose data, his right').

Q29. Which US-led initiative did India formally join on 20 February 2026?

(A) Artemis Accords
(B) Five Eyes
(C) Pax Silica
(D) Quad Chip Pact

Answer: (C) Pax Silica. Pax Silica - a US-led technology alliance.

Q30. The IndiaAI Mission has a budget outlay of:

(A) Rs 50,000 crore
(B) Rs 10,000 crore
(C) Rs 5,000 crore
(D) Rs 1,000 crore

Answer: (B) Rs 10,000 crore. About Rs 10,000 crore.

Q31. Which of the following is an Indian sovereign AI model unveiled in February 2026?

(A) BharatGen Param2
(B) ChatGPT
(C) Llama
(D) Gemini

Answer: (A) BharatGen Param2. BharatGen's Param2 (also Sarvam AI's LLMs and Gnani AI's Vachana voice stack).

Q32. India Semiconductor Mission 2.0 was announced in the:

(A) Monetary Policy Oct 2026
(B) AI Summit declaration
(C) Economic Survey 2025-26
(D) Union Budget 2026-27

Answer: (D) Union Budget 2026-27. Union Budget 2026-27.

Q33. ★ The Nipah virus is mainly carried by:

(A) Ticks
(B) Fruit bats
(C) Mosquitoes
(D) Sandflies

Answer: (B) Fruit bats. Fruit bats; also transmitted via pigs or contaminated fruit.

Q34. The Nipah cases reported in late December 2025 were in which state?

(A) Karnataka
(B) Kerala
(C) West Bengal
(D) Assam

Answer: (C) West Bengal. West Bengal - two health workers (India's 7th Nipah outbreak).

Q35. The fatality rate of Nipah virus infection ranges roughly between:

(A) 1-5%
(B) 10-20%
(C) 40-75%
(D) 90-99%

Answer: (C) 40-75%. 40-75%; there is no licensed cure or vaccine yet.

Q36. ★ 'AMR' in health stands for (asked in OAVS TGT English 2019):

(A) Anti-Microbial Resistance
(B) Advanced Medical Research
(C) Antibody Mapping Report
(D) Adult Mortality Rate

Answer: (A) Anti-Microbial Resistance. Anti-Microbial Resistance.

Q37. 5G services were launched in India on 1 October 2022 during the _____ edition of the India Mobile Congress.

(A) 4th
(B) 5th
(C) 6th
(D) 7th

Answer: (C) 6th. 6th (asked in OAVS TGT Social Studies 2023).

4. Rapid-Revision Table

Event	Answer
Gaganyaan G1 (uncrewed) / robot	**~Dec 2026 on HLVM3 / Vyommitra; crewed flight 2027**
ISRO Chairman	**Dr V. Narayanan**
NISAR launch	**30 Jul 2025, GSLV-F16 (ISRO-NASA)**
Axiom-4 / SpaDeX	**Shubhanshu Shukla at ISS (2025) / docking 16 Jan 2025 - 4th country**
EOS-05	**GSLV-F17, 4 Sep 2026 - first launch after 7-month gap**
Artemis II	**Splashdown 11 Apr 2026; crewed landing planned 2028**
Ramsar sites	**103 (Sukhna, Sauj Jheel added on 15 Oct 2026)**
World Heritage - India	**45 (Sarnath, 48th session, Busan, July 2026); 44th Maratha Military Landscapes**
Tiger estimation	**2022: 3,682; AITE 2026 report due 2027; 58 reserves**
Cheetahs	**52 (Kuno 49, Gandhi Sagar 3); 9 from Botswana (Feb 2026)**
India's NDC 2031-35	**47% intensity cut (vs 2005); ~60% non-fossil capacity; net-zero 2070**
AI Impact Summit	**Bharat Mandapam, Feb 2026; MANAV vision; Pax Silica (20 Feb)**
Nipah	**West Bengal, Dec 2025 (7th outbreak); fruit bats; 40-75% fatal**
PYQ anchors	**TX2 - Pench; NCAP - MoEFCC; 5G - 1 Oct 2022; AMR = Anti-Microbial Resistance**

Check before exam: Gaganyaan G1 (due about December 2026), any new Ramsar/World Heritage additions and COP31 outcomes will change after 28 Oct 2026 - re-check them close to the exam.

Sources

• UNI India, Free Press Journal, India TV, The Week - Gaganyaan, NISAR, Axiom-4, GSLV-F17/EOS-05
• Chronicle India, Utkarsh, Superkalam - India's Ramsar sites reach 103
• Drishti IAS, Manorama Yearbook - Sarnath, 48th World Heritage Committee session
• GKToday, Adda247 - All India Tiger Estimation 2026 and International Tiger Day 2026; Shankar IAS - cheetahs in India 2026
• ThePrint, Adda247 - India's NDC 2031-35
• Free Press Journal - India AI Impact Summit 2026 (MANAV, Pax Silica); Digital India press releases - ISM 2.0
• Dawn, Business Standard, WHO - Nipah in West Bengal; ThePrint - Artemis II